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20 / 20 submitted

Q1. Void Function (No Return Value)

CODE

```
def welcome():  
    print("Welcome to Python Programming")  
  
welcome()
```

OUTPUT (no output)

Q2. Void Function (No Return Value)

CODE

```
def table(n):  
    for i in range(1, 11):  
        print(n, "x", i, "=", n * i)  
  
n = int(input())  
table(n)
```

INPUT 5

OUTPUT (no output)

Q3. Void Function (No Return Value)

CODE

```
def print_even(n):  
    for i in range(2, n + 1, 2):  
        print(i)  
  
n = int(input())  
print_even(n)
```

INPUT 10

OUTPUT (no output)

Q4. Fruitful Function (Returns One Value)

CODE

```
def count_vowels(s):  
    count = 0  
    vowels = "aeiouAEIOU"  
    for ch in s:  
        if ch in vowels:  
            count += 1  
    return count  
  
s = input()  
print(count_vowels(s))
```

INPUT Hello World

OUTPUT (no output)

Q5. Fruitful Function (Returns One Value)

CODE

```
def largest_element(lst):  
    return max(lst)  
  
lst = list(map(int, input().split()))  
  
print(largest_element(lst))
```

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INPUT

10 25 8 40 15

OUTPUT

(no output)

Q6. Fruitful Function (Returns One Value)

CODE

```
def is_prime(n):  
    if n ≤ 1:  
        return "Not Prime"  
    for i in range(2, int(n ** 0.5) + 1):  
        if n % i == 0:  
            return "Not Prime"  
    return "Prime"  
  
n = int(input())  
print(is_prime(n))
```

INPUT

7

OUTPUT

(no output)

Q7. Function Returning Two or More Values

CODE

```
def student(m1, m2, m3):  
    total = m1 + m2 + m3  
    average = total / 3  
  
    if average ≥ 90:  
        grade = "A"  
    elif average ≥ 75:  
        grade = "B"  
    elif average ≥ 50:  
        grade = "C"  
    else:  
        grade = "F"  
  
    return total, average, grade  
  
m1 = int(input())  
m2 = int(input())  
m3 = int(input())  
  
t, a, g = student(m1, m2, m3)  
  
print(t)  
print(a)  
print(g)
```

INPUT

80
90
70

OUTPUT

(no output)

Q8. Function Returning Two or More Values

CODE

```
import math

def circle(radius):
    area = math.pi * radius * radius
    circumference = 2 * math.pi * radius
    return area, circumference

r = float(input())

a, c = circle(r)

print("Area:", a)
print("Circumference:", c)
```

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INPUT

5

OUTPUT

(no output)

Q9. Function Returning Two or More Values

CODE

```
import math

def calculate(num):
    square = num ** 2
    cube = num ** 3
    square_root = math.sqrt(num)
    return square, cube, square_root

n = float(input())

sq, cu, sr = calculate(n)

print("Square:", sq)
print("Cube:", cu)
print("Square Root:", sr)
```

INPUT

4

OUTPUT

(no output)

Q10. Keyword Arguments

CODE

```
def student(name, age, department):
    print("Name:", name)
    print("Age:", age)
    print("Department:", department)

name = input()
age = int(input())
department = input()

student(name=name, age=age, department=department)
```

INPUT

Ramya
20
Bioinformatics

OUTPUT

(no output)

Q11. Keyword Arguments

CODE

```
def employee(empid, name, salary):  
    print("Employee ID:", empid)  
    print("Name:", name)  
    print("Salary:", salary)  
  
empid = input()  
name = input()  
salary = input()  
  
employee(salary=salary, name=name, empid=empid)
```

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INPUT

101
Ramya
50000

OUTPUT

(no output)

Q12. Keyword Arguments

CODE

```
def book(title, author, price):  
    print("Title:", title)  
    print("Author:", author)  
    print("Price:", price)  
  
title = input()  
author = input()  
price = input()  
  
book(price=price, title=title, author=author)
```

INPUT

Python Basics
Guido
499

OUTPUT

(no output)

Q13. Variable-Length Arguments (*args)

CODE

```
def find_sum(*args):  
    total = 0  
    for i in args:  
        total += i  
    return total  
  
nums = list(map(int, input().split()))  
print(find_sum(*nums))
```

INPUT

10 20 30 40

OUTPUT

(no output)

Q14. Variable-Length Arguments (*args)

CODE

```
def average(*args):  
    return sum(args) / len(args)  
  
marks = list(map(int, input().split()))  
  
print(average(*marks))
```

INPUT

80 90 70

OUTPUT

(no output)

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**Q15. Variable-Length Arguments (*args)**

CODE

```
def print_strings(*args):  
    for s in args:  
        print(s)  
  
n = int(input())  
strings = []  
  
for i in range(n):  
    strings.append(input())  
  
print_strings(*strings)
```

INPUT

3
Apple
Banana
Mango

OUTPUT

(no output)

Q16. Variable-Length Arguments (*args)

CODE

```
def largest(*args):  
    return max(args)  
  
n = int(input())  
nums = []  
  
for i in range(n):  
    nums.append(int(input()))  
  
print(largest(*nums))
```

INPUT

5
10
25
7
42
18

OUTPUT

(no output)

Q17. Recursive Functions

CODE

```
def fibonacci(n):  
    if n ≤ 1:  
        return n  
    return fibonacci(n - 1) + fibonacci(n - 2)  
  
n = int(input())  
  
for i in range(n):  
    print(fibonacci(i), end=" ")
```

INPUT 7

OUTPUT (no output)

Q18. Recursive Functions

CODE

```
def sum_digits(n):  
    if n == 0:  
        return 0  
    return n % 10 + sum_digits(n // 10)  
  
n = int(input())  
print(sum_digits(n))
```

INPUT 12345

OUTPUT (no output)

Q19. Recursive Functions

CODE

```
def reverse_string(s):  
    if len(s) == 0:  
        return s  
    return reverse_string(s[1:]) + s[0]  
  
s = input()  
print(reverse_string(s))
```

INPUT Hello

OUTPUT (no output)

Q20. Recursive Functions

CODE

```
def book(title, author, price):  
    print("Title:", title)  
    print("Author:", author)  
    print("Price:", price)  
  
title = input()  
author = input()  
price = input()  
  
book(author=author, price=price, title=title)
```

INPUT Python Basics
John
450

OUTPUT (no output)